

Terms of Reference

**Hiring of Agency for Design, Development,
Implementation & Comprehensive Operations and Maintenance
of the Centralised Online Portal for Solid Waste Management
under SWM Rules, 2026**



Central Pollution Control Board (CPCB)

Ministry of Environment, Forest and Climate Change

April 2026

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1 Background

The Central Pollution Control Board (CPCB), under the Ministry of Environment, Forest and Climate Change (MoEFCC), is the apex regulatory body responsible for the prevention and control of pollution in India. The Solid Waste Management Rules, 2026, gazette-notified and effective from 01.04.2026, mandate a comprehensive overhaul of India's solid waste governance framework, transitioning to a digital-first, circular economy model.

The SWM Rules, 2026 require every local body (Urban Local Body / Rural Local Body), Bulk Waste Generator (BWG), waste processor, and facility operator to register on a Centralised Online Portal managed by CPCB. The portal is intended to enable registration, compliance monitoring, returns filing, certificate management under the Extended Bulk Waste Generator Responsibility (EBWGR) framework, environmental compensation administration, and data-driven decision-making.

CPCB intends to engage a qualified System Integrator (SI) through NICS I empanelment for the end-to-end design, development, implementation, and comprehensive operations and maintenance of the Centralised SWM Portal. The Agile methodology shall be adopted for portal development to ensure iterative delivery, stakeholder feedback integration, and compliance with the accelerated timeline.

Registration fees and other payments collected through the portal shall be shared with the concerned State Pollution Control Board (SPCB) / Pollution Control Committee (PCC) as per the revenue-sharing percentage determined and notified by CPCB from time to time. The SI shall implement the necessary payment splitting, reconciliation, and disbursement mechanisms to facilitate this revenue-sharing arrangement.

2 Objective

The primary objective of this engagement is to design, develop, deploy, and maintain a national-scale, secure, and scalable Centralised Online Portal for Solid Waste Management, enabling CPCB to:

- Facilitate online registration of all ULBs, RLBs, Bulk Waste Generators (BWGs), Waste Processing Facilities, MRFs, Sanitary Landfills, and other regulated entities as mandated under SWM Rules, 2026
- Enable end-to-end compliance monitoring and regulatory oversight across all tiers: CPCB, SPCBs/PCCs, District Administration, and Local Bodies
- Implement the EBWGR certificate lifecycle including generation, procurement, transfer, tracking, and marketplace operations (integration with Electronic Trading Platform (ETP) via API)
- Support filing of annual returns, monthly/weekly/daily waste reports, and periodic compliance submissions by all registered entities
- Administer Environmental Compensation calculations, escrow fund management, and enforcement workflows
- Provide real-time dashboards, analytics, and reporting for national, state, district, and ULB-level monitoring
- Implement revenue-sharing mechanisms for distributing registration fees and other collections to SPCBs/PCCs as per CPCB-notified percentages
- Integrate with Government platforms including LGD Master Data, Aadhaar eKYC, UCAMS (Common Consent Management), Common EPR Portal, GOBARDHAN, National Single Window System (NSWS), and the Payment Gateway (PG) as selected by CPCB
- Support mobile inspection workflows for ULBs/SPCBs and grievance redressal mechanisms for all stakeholders

The vendor shall undertake comprehensive requirement understanding, design, development, implementation, and operations and maintenance of the SWM Portal, including all modules described in Section 3.

3 Brief Overview

The Centralised SWM Portal is a web-based, multi-tenant platform to be hosted on National Government Cloud (or Meity empanelled cloud service provider). It shall serve as the single national repository for all solid waste management registrations, compliance data, returns, certificates, and enforcement actions under SWM Rules, 2026. The portal shall be accessible to multiple stakeholder categories including BWGs, ULBs/RLBs, SPCBs/PCCs, waste processing facilities, CPCB officials, District Collectors, and the public.

3.1 Features of Core Modules

- a) Entity Registration & Onboarding:** Self-registration workflows for ULBs/RLBs, BWGs (residential, commercial, government/PSU, social infrastructure categories), Waste Processing Facilities, MRFs, and Sanitary Landfill operators. Multi-step forms with conditional logic, document upload, and fee payment via the Payment Gateway (PG) as selected by CPCB.
- b) Authentication & Access Management:** OTP-based or DSC-based login, PAN/CIN/GSTIN verification (if required), role-based access control (RBAC) for all portal user types.
- c) BWG Registration & Compliance:** BWG registration workflow (self-registration, ULB-initiated), wet waste processing strategy capture (on-site/vendor tie-up), MoHUA per-capita waste generation norms engine, registration certificate generation.
- d) EBWGR Certificate Management:** Generation, procurement, transfer, and lifecycle tracking of Extended Bulk Waste Generator Responsibility certificates. Includes integration with marketplace for certificate trading (ETP platform).
- e) Returns & Reporting Engine:** Annual returns (Form III, Form IV, Form VI), monthly waste reports by ULBs, quarterly compliance submissions, and scheduled report generation with PDF/CSV/Excel export.
- f) Compliance Monitoring & Dashboards:** National, state, district, and ULB-level dashboards with real-time compliance indicators, registration counts, waste processing metrics, and EBWGR status.
- g) Payment & Revenue Management:** Registration fee collection via the PG selected by CPCB, EBWGR certificate charges, Environmental Compensation escrow fund, late payment surcharge computation, and SPCB revenue-sharing disbursement.
- h) Mobile Inspection App:** Mobile-based (Android/iOS) inspection workflows for ULB and SPCB field officers with photo/video upload, geo-tagging, checklist-based inspections, and offline mode.
- i) Audit Management:** Periodic audit scheduling for all entity types, audit checklist generation, findings recording, and corrective action tracking.
- j) Grievance Redressal Portal:** Grievance submission by all stakeholders (BWGs, ULBs, facility operators, and the public), tracking, escalation workflows, and resolution monitoring.
- k) GIS & Spatial Analytics:** Geo-mapping of registered entities, MRF/landfill/facility locations, ward-level coverage analysis using PostGIS or open-source/Government GIS tools (Bharat Maps / Bhuvan).

3.2 Modules under the SWM Portal

The portal comprises the following 24 modules across four development phases:

Sl#	Phase	Module / Component	Brief Scope (Indicative in nature and may change as per project requirement)
1	Phase 1	Admin Seeding, State Hierarchy & RBAC	LGD API import of ULBs/GPs; State admin account creation; Nodal Officer designation; ULB Monitoring Cell role setup; BWG bylaw upload; User charge and penalty schedule configuration; Role-based access control for Admin, Applicant, Reviewer, SPCB, ULB Officer, and Monitoring Cell roles.
2	Phase 1	User Authentication & Account Management	OTP / DSC login; PAN/Aadhaar/CIN/GSTIN API verification; session management; password/OTP recovery; multi-factor authentication.
3	Phase 1	ULB / RLB Self-Registration	ULB/RLB profile creation: LGD Code auto-fill, ward count, SBM-U/AMRUT ID, nodal officer, SPCB region mapping, Gazette notification upload
4	Phase 1	BWG Registration (Workflow + Fee)	Self-registration, ULB-initiated, and public notice tracks; entity category and sub-type taxonomy; per capita auto calculation norms (MoHUA Annexure-7); non-BWG self-declaration; fee slab auto calculation; Payment Gateway integration; document vault; RWA application; event intimation sub-form; MoHUA norms and Fee Calculator.
5	Phase 1	Facility Registration + Fee	Waste Processing Facility, RDF Plant, MRF, Sanitary Landfill, CBG/Biogas plant operator registration; facility type taxonomy; lat-long capture; TPD-based fee slab; GOBARDHAN ID linkage; linked ULB jurisdiction.
6	Phase 1	Document Vault, Upload Engine & Certificate Generation	Multi-format upload (PDF/JPG/PNG); virus scan; document type tagging; version management; issuance of registration certificate with system-generated URN; QR code; digital signature; certificate download and verification.
7	Phase 1	Payment Gateway & SPCB Revenue Sharing	Registration fee collection via the PG selected by CPCB; fee slab engine; late payment surcharge; real-time payment splitting between CPCB and SPCB/PCC as per CPCB-notified revenue-sharing percentage; reconciliation dashboard; payment receipt generation.
8	Phase 1	Registration Dashboard	National, state, and district-level registration count dashboard; entity-type filters; real-time counters for SC affidavit compliance reporting; SPCB/PCC-specific view.
9	Phase 2	ULB Physical Verification & Final Certificate	ULB officer verification of BWG premises; 3-day verification protocol; weighment recording; de-listing workflow; issuance of final registration certificate post-verification; certificate validity and renewal workflows.
10	Phase 2	Operations, Returns & Compliance Management	Annual returns (Form III, IV, VI) by ULBs, BWGs, and facility operators; monthly waste reports by ULBs; quarterly compliance submissions; return filing engine with validation rules; compliance score computation; automated non-compliance alerts; three-tier enforcement trigger.
11	Phase 2	EBWGR Certificate Lifecycle & Marketplace	EBWGR certificate generation, procurement, transfer, and lifecycle tracking; buyer-seller marketplace for certificate trading; price discovery; transaction history; linked to BWG compliance status.
12	Phase 2	Payment Management (Extended) & EC Engine	EBWGR certificate charges; Environmental Compensation calculation; escrow fund management; fund disbursement; service payments; SPCB revenue-share reconciliation via the PG selected by CPCB.
13	Phase	Appeal Management	Three-tier appeal workflows for registration rejection,

Sl#	Phase	Module / Component	Brief Scope (Indicative in nature and may change as per project requirement)
	2		compliance penalties, and EBWGR disputes; Reporting data edit, SLA-based escalation; disposition tracking.
14	Phase 2	Audit Management	Periodic audit scheduling for all entity types; audit checklist generation; findings recording; corrective action tracking; compliance gap analysis reports.
15	Phase 2	Sanitary Landfill Management	Landfill site registration; operational data capture; legacy waste dumpsite registry; bio-remediation tracking; capacity monitoring; closure and post-closure monitoring provisions.
16	Phase 2	SPCB/PCC Dashboard, Workflow & ULB Services	SPCB/PCC officer dashboards; application review and approval workflows; state-level compliance monitoring; consent/authorization management (linked to UCAMS); ULB service configuration; SWM hilly/special areas provisions.
17	Phase 2	Grievance Redressal Portal	Grievance submission by all stakeholders (BWGs, ULBs, facility operators, SPCBs, and the public); tracking dashboard; escalation to ULB/SPCB/CPCB; resolution SLA monitoring; grievance analytics.
18	Phase 3	Advanced Returns, Reporting & Analytics	Comprehensive return filing for all entity types; automated validation; 10+ internal and 10+ external report templates; trend analysis; state performance ranking; scheduled report generation; PDF/CSV/Excel export.
19	Phase 3	External System Integrations	API integrations with UCAMS (Common Consent Management), Common EPR Portal, GOBARDHAN, NSWs, SBM-U portal, LGD; inter-portal data exchange; API gateway for state portals.
20	Phase 3	Mobile Inspection App	Field inspection mobile app (Android/iOS) for ULB and SPCB officers; photo/video capture; geo-tagging; checklist-based inspection workflows; offline mode with sync; inspection report generation.
21	Phase 3	GIS Module & Spatial Analytics	Geo-mapping of all registered entities; ward-level coverage analysis; facility catchment area mapping; PostGIS with open-source/Government GIS tools (Bharat Maps / Bhuvan); spatial dashboards.
22	Phase 4	AI/ML Analytics & Predictive Compliance	Anomaly detection in waste data; compliance pattern analysis; predictive non-compliance scoring; data quality flagging; operational insights dashboards.
23	Phase 4	Notifications, Communication & Air Quality Monitoring	SMS/email/push notification management; configurable templates; bulk communication; regulatory notice/circulars workflows; AQI data capture at landfill sites; sensor integration or manual entry; threshold-based alerts.
24	Phase 4	Carbon Value Realisation (Phase-gated)	Carbon credit MRV framework; project registry; credit calculation engine; dashboard. Scope to be confirmed with MoEFCC before development commences.

3.3 Indicative Technology Stack

The following technology stack is indicative in nature and may be adopted for the SWM portal development:

Sl#	Component	Technology
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a)	Hosting	NGC / MeitY empanelled Government Cloud
b)	Backend Framework	Node.js / Springboot
c)	Frontend Framework	Angular/React
d)	Database	PostgreSQL (primary RDBMS)

3.4 Scale of the System

a) Approximately 50,00,000 (fifty lakh) registered users are expected across all entity types over the portal lifecycle, including ULBs/RLBs, BWGs (across residential, commercial, government/PSU, and social infrastructure categories, gated communities and institutions), waste processing facilities, MRFs, landfill operators, SPCB/PCC officials, and CPCB users.

b) Over 5,000 ULBs and 2,50,000+ Gram Panchayats are expected to register on the portal for compliance with SWM Rules, 2026.

c) BWG registrations are expected to number 25,00,000+ entities, spanning hotels, restaurants, residential complexes, commercial establishments, government buildings, educational institutions, hospitals, and event venues.

d) The system shall handle concurrent filing of annual returns, monthly waste reports, and quarterly compliance submissions from all registered entities.

e) The portal shall support integration with multiple Government platforms including LGD, Aadhaar, UCAMS (Common Consent), Common EPR Portal, GOBARDHAN, NSWIS, and the PG selected by CPCB.

f) Database size is expected to grow to approximately 20 TB or more over the first three years, considering registration data, returns, certificates, documents, GIS data, and EBWGR transaction records.

g) The payment engine shall support SPCB-wise revenue splitting for all fee collections, with real-time reconciliation across 28 States and 8 Union Territories.

4 Scope of Work

The selected System Integrator shall be responsible for end-to-end design, development, implementation, and comprehensive operation and maintenance of the Centralised SWM Portal, including but not limited to the following:

4.1 Requirement Gathering & Analysis

The SI shall be solely responsible for comprehensive requirement gathering and analysis for all modules of the SWM Portal. CPCB and relevant CPCB divisions shall provide domain inputs, regulatory guidelines, and approvals:

- Prepare UI/UX wireframes/prototypes for all key workflows (registration forms, dashboards, certificate generation, returns filing, grievance submission) for CPCB review before development.
- Finalise phase wise scope with CPCB at the beginning of the phase.
- Prepare and maintain a comprehensive Test Plan for each phase.
- Prepare User Manuals (separate for each user role: BWG, ULB, SPCB, CPCB Admin, Facility Operator) and Administrator Manuals for each phase go-live.
- Prepare a Training Plan and conduct training sessions for CPCB officials, SPCB officers, and ULB users for each phase go-live. Training materials (presentations and quick-reference guides) shall be provided.

4.2 Design and Development